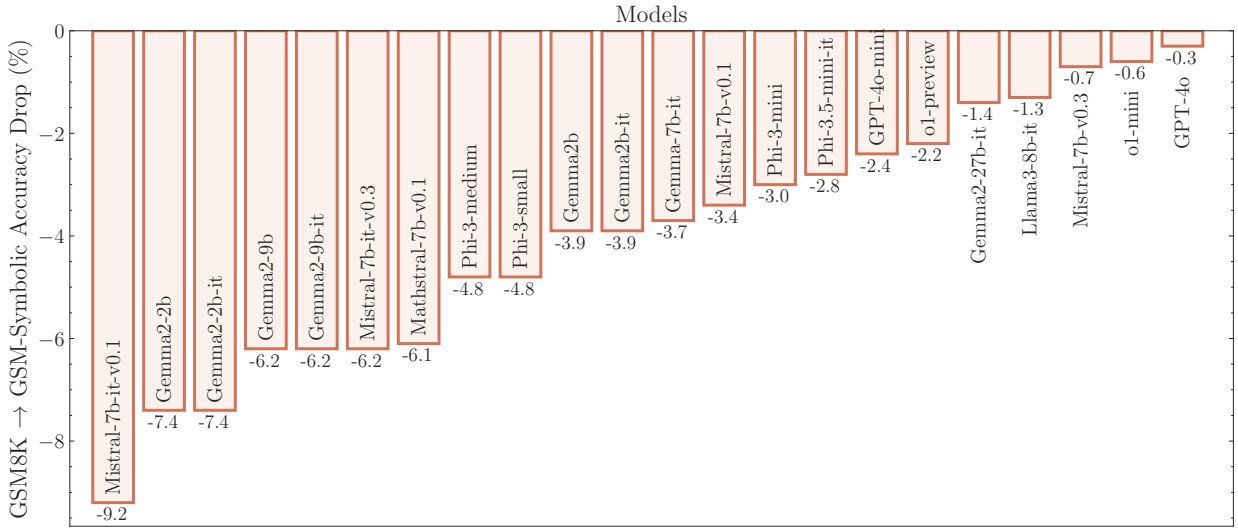

Figure 3: The performance of all state-of-the-art models on `GSM-Symbolic` drops compared to `GSM8K`. Later, we investigate the factors that impact the performance drops in more depth.

Another noteworthy observation is that the performance (represented by the dashed line in Fig. 2) on the original questions from the 100 examples of `GSM8K` used as templates is often more than one standard deviation away from the center of the `GSM-Symbolic` performance distribution, frequently on the *right side* of the distribution (this holds for 21 out of 25 models). One explanation for this could be data contamination, where some of the test examples from `GSM8K` inadvertently ended up in the training set of these models, leading to an optimistic bias in performance. Fig. 3 shows the performance drop from `GSM8K` to `GSM-Symbolic` for several models. We can see that for models such as Gemma2-9B, Phi-3, Phi-3.5, and Mathstral-7B, the dashed line in Fig. 2 lies on the right side, and the drop in performance is higher than for models such as Llama3-8b and GPT-4o, where the performance on `GSM8K` is close to the center of the `GSM-Symbolic` distribution and the drop in performance is negligible. In Appendix A.3, we present further results to support this claim for other models such as Phi-2 and Mistral-7B. These results lead us to investigate the fragility of the reasoning abilities of LLMs in the next section.

4.2 How Fragile is Mathematical Reasoning in Large Language Models?

In the previous sub-section, we observed high performance variation across different sets generated from the same templates, along with a performance degradation compared to the original GSM8K accuracy. This suggests that the perceived reasoning process of language models may not be formal and is hence susceptible to changes. One explanation is that these models attempt to perform a kind of in-distribution pattern-matching, aligning given questions and solution steps with similar ones seen in the training data. As no formal reasoning is involved in this process, it could lead to high variance across different instances of the same question. In this sub-section and the next one, we investigate these observations further and we show that several factors contribute to the performance variation of the models.

First, we investigate the impact of the *type* of change to understand the difference between changing names (e.g., person names, places, foods, currencies, etc.) versus changing numbers (i.e., the values of variables).

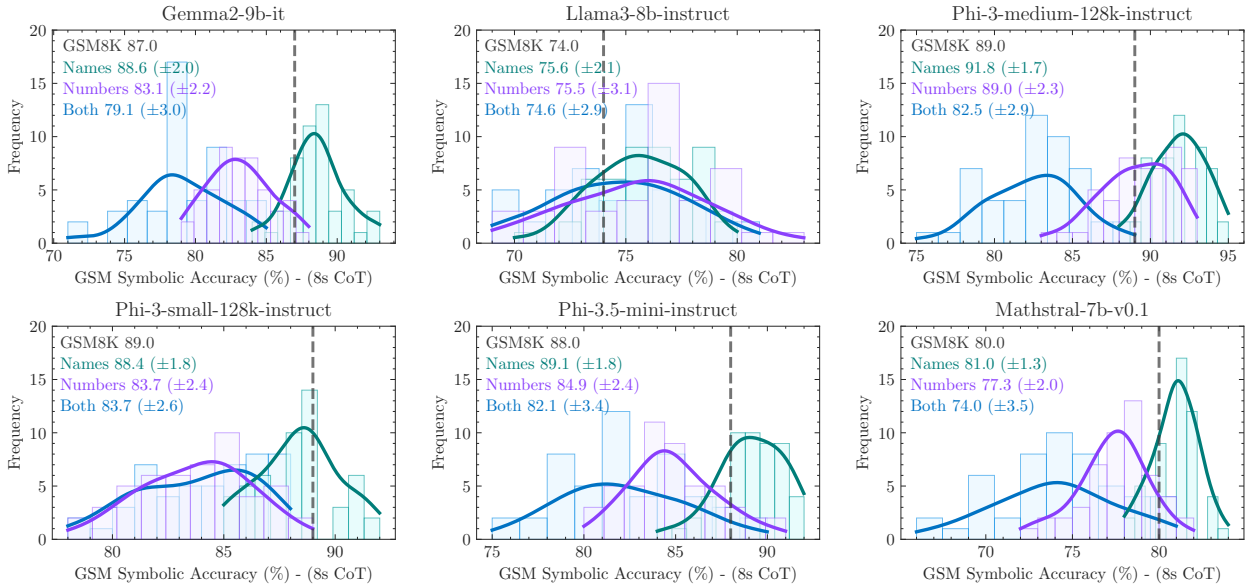

Figure 4: How sensitive are LLMs when we change only names, only proper numbers, or both names and numbers? Overall, models have noticeable performance variation even if we only change names, but even more when we change numbers or combine these changes.

Figure 4 demonstrates that while performance variation persists, the variance is lower when changing names compared to numbers. Notably, the original `GSM8K` accuracy of models is now much closer to the center of the changed proper names distribution, in contrast to changed numbers or both. Furthermore, a gradual shift in the means of distributions from right to left, along with an increase in variance, is evident across almost all models. It is both striking and concerning that such performance variance exists when only changing proper names, as this level of variability would not be expected from a grade-school student with genuine mathematical understanding.

From the results in this section, we observe that by increasing the *difficulty* of changes (from names to numbers), the performance drops and the variance increases, overall suggesting that the reasoning capabilities of state-of-the-art LLMs are fragile for the aforementioned reasons. Assuming that LLMs are not performing formal reasoning, how important is the question *difficulty* on the distribution of performance? In the next section, we study this question further.

4.3 How Does Question Difficulty Affect Performance Distribution?

The results in the previous subsection motivate us to study the impact of question *difficulty* on the mean and variance of the performance distribution. To this end, we generate several new templates from the `GSM-Symb`, as illustrated in Fig. 5. First, by removing one clause, we obtain `GSM-Symbolic-Minus-1` or `GSM-M1` for short. Similarly, we can add one or two clauses to the questions to increase the difficulty, resulting in `GSM-Symbolic-Plus-1` (`GSM-P1`) and `GSM-Symbolic-Plus-2` (`GSM-P2`), respectively[1].

[1]It is important to recognize that adding or removing a clause does not always result in an exact increase or decrease of one in the number of required reasoning steps. In general, the exact number of steps needed to solve a problem is not fixed, as there may be multiple valid solutions for each problem, each requiring a different number of steps. Regardless, our main focus in this section is to understand the *evolution* of the performance distribution rather than the precise performance metrics.

Different Levels of GSM-Symbolic Difficulty

GSM-Symbolic-M1: To make a call from a phone booth, you must pay $0.6 for each minute of your call. ~~After 10 minutes, that price drops to $0.5 per minute~~.How much would a 60-minute call cost?

GSM-Symbolic: To make a call from a phone booth, you must pay $0.6 for each minute of your call. After 10 minutes, that price drops to $0.5 per minute. How much would a 60-minute call cost?

GSM-Symbolic-P1: To make a call from a hotel room phone, you must pay $0.6 for each minute of your call.After 10 minutes, that price drops to $0.5 per minute. After 25 minutes from the start of the call, the price drops even more to $0.3 per minute.How much would a 60-minute call cost?

GSM-Symbolic-P2: To make a call from a hotel room phone, you must pay $0.6 for each minute of your call. After 10 minutes, the price drops to $0.5 per minute. After 25 minutes from the start of the call, the price drops even more to $0.3 per minute. If your total bill is more than $10, you get a 25% discount. How much would a 60-minute call cost?

Figure 5: Modifying the difficulty level of GSM-Symbolic by modifying the number of clauses.

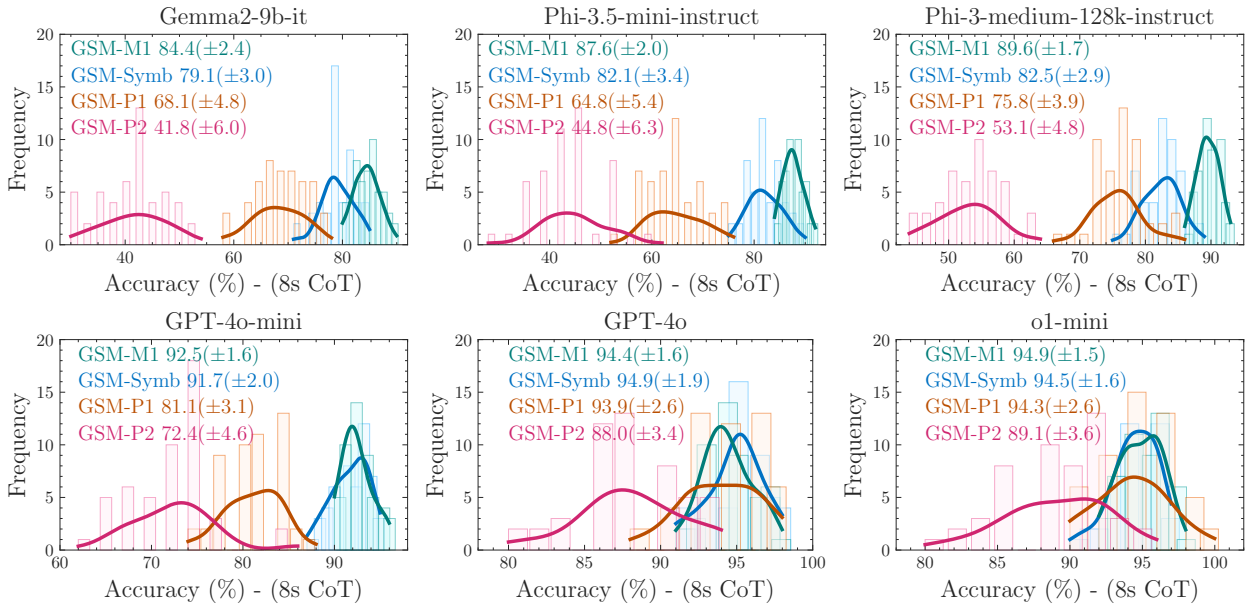

Figure 6: The impact of increasing the number of clauses on performance: As the difficulty increases from GSM-M1→ GSM-Symb→ GSM-P1→ GSM-P2, the distribution of performance shifts to the left (i.e., accuracy decreases), and the variance increases.

As shown in Fig. 6, the trend of the evolution of the performance distribution is very consistent across all models: as the difficulty increases, the performance decreases and the variance increases. Note that overall, the *rate of accuracy drop* also increases as the difficulty increases. This is in line with the hypothesis that models are not performing formal reasoning, as the number of required reasoning steps increases linearly, but the rate of drop seems to be faster. Moreover, considering the pattern-matching hypothesis, the increase in variance suggests that searching and pattern-matching become significantly harder for models as the difficulty increases.